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Simple and robust near-infrared spectroscopic monitoring of indium-tin-oxide (ITO) etching solution using Teflon tubing

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Abstract

The ability to monitor etching solutions using a spectroscopy directly through existing Teflon lines in electronic industries is highly beneficial and offers many advantages. A monitoring method was developed using near-infrared (NIR) measurements with Teflon tubing as a sample container for the quantification of components in the indium-tin-oxide (ITO) etching solution composed of hydrochloric acid (HCl), acetic acid (CH_3COOH) and water. Measurements were reproducible and it was possible to use the same calibration model for different Teflon tubings. Even though partial least squares (PLS) calibration performance was slightly degraded for Teflon cells when compared to quartz cells of the similar pathlength, the calibration data correlated well with reference data. The robustness of Teflon-based NIR measurement was evaluated by predicting the spectra of 10 independent samples that were collected using five different Teflon tubes. Although, two Teflon tubes were visually less transparent than the other three, there was no significant variation in the standard error of predictions (SEPs) among the five Teflon tubes. Calibration accuracy was successfully maintained and highly repeatable prediction results were achieved. This study verifies that a Teflon-based NIR measurement is reliable for the monitoring of etching solutions and it can be successfully integrated into on-line process monitoring.
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Keywords: Near-infrared spectroscopy; Indium-tin-oxide (ITO) etching solution; Teflon

1. Introduction

Previous study demonstrated the ability to measure the important components of an etching solution, including HF, using near-infrared (NIR) spectroscopy [1,2] that employs Teflon tubing (perfluoroalkoxy (PFA) fluorocarbon) as a sample cell [3]. Direct NIR spectroscopic measurement without any process system modifications would be beneficial as Teflon tubing is currently being used at wet stations. Additionally, Teflon is inert to the major components of etching solutions including acids such as HF [4]. The previous study, which was based on proof of principle, did not consider other parameters such as signal-to-noise ratio, prediction stability with different Teflon tubes and robustness of calibration that are key to determine the feasibility of this methodology in real industrial applications. The goal of this research is to investigate the above parameters and demonstrate the practical applicability of an on-line Teflon-based NIR measurement. Calibration performance was investigated for the measurement of the components in the indium-tin-oxide (ITO) etching solution, which contains hydrochloric acid (HCl), acetic acid (CH_3COOH) and water. ITO solution was selected as one possible candidate for the proposed unique Teflon-based measurement. This etching solvent is used widely in industry and real-time, on-line monitoring is essential for optimal process operation [5]. Initially, we have evaluated the possible degradation of signal-to-noise ratio due to the use of optical fiber configuration (shown in Fig. 1 (configuration B)) by examining root mean square (RMS) noise of spectra collected. Additionally, as a comparison, three rectangular quartz cells with pathlengths close to the Teflon tubes were also used to collect spectra of the same samples. Because the optical quality of Teflon is not necessarily as consistent as quartz or CaF_2, it was necessary to

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investigate possible variations in prediction performance which are critical in real industrial settings when different Teflon tubes may be used.

To have reliability of the Teflon-based measurement in real industry, the prediction accuracy should be maintained even when different Teflon tubes are used. If it is not realized, calibration models must be developed for each Teflon cell used, which is practically difficult and unrealistic. The prediction stability was evaluated by predicting the sample spectra collected by five different Teflon tubes. Finally, prediction repeatability of the Teflon-based measurement was investigated.

2. Experimental

2.1. Sample preparation

Twenty-eight standard samples were prepared by mixing appropriate amounts of HCl, acetic acid and water. The concentration of each component was determined by generating 28 random numbers in a given concentration range. The concentrations of HCl, acetic acid and water ranged from 18.6 to 31.3, 4.8 to 14.5 and 58.5 to 74.0 wt.%, respectively. The concentration ranges of the standard mixtures were slightly wider than the actual concentration ranges in the process. The chemicals were purchased from a common supplier. Each sample was prepared just prior to NIR spectral collection.

2.2. NIR spectra and data processing

NIR spectra were collected using two different schemes initially, as shown in Fig. 1. A half inch Teflon tube (outer diameter: 12.7 mm, inner diameter: 9.68 mm) was positioned in the beam path, so the focused NIR beam (approximately 4 mm × 3 mm in size) directly penetrated the center of Teflon tube (referred to as configuration A). An ABB Bomem MB-160 FT-NIR spectrometer (Quebec City, Quebec, Canada) equipped with a tungsten–halogen source and DTGS detector was used for configuration A. In a second scheme, NIR spectra were collected by an optical fiber configuration as described in Fig. 1 (configuration B). NIR radiation from the optical fiber was illuminated perpendicularly to the center of Teflon tube and the transmitted radiation was then reflected and focused into the receiving optical fiber. An ABB Bomem fiber optic-based FT-NIR spectrometer equipped with a tungsten–halogen source and a Germanium detector was used for configuration B. All NIR spectra from both configurations corresponded to an accumulation of 128 scans with a resolution of 32 cm^{-1} (acquisition time: approximately 1 min). NIR spectra were collected at room temperature immediately after sample preparation to minimize evaporation. All spectral calculations including partial least squares (PLS) regression and second derivative algorithms were accomplished using Matlab version 6.5 (MathWorks Inc., MA, USA).

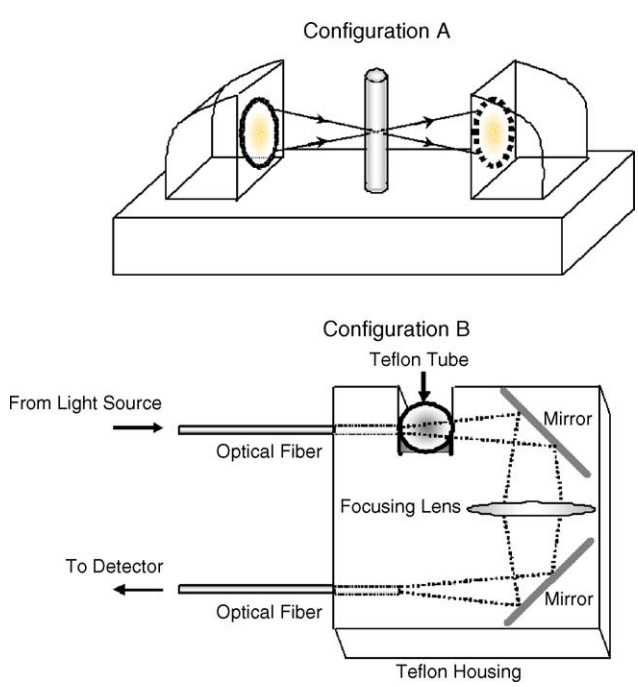

Fig. 1. The schematic diagram of two different configurations for spectral collection using Teflon tubing.

3. Results and discussion

3.1. Signal-to-noise ratio of the optical fiber configuration

Fig. 2 shows the single beam spectra of air and two blanks from an empty Teflon tube that were simultaneously collected by configuration A via direct transmission and configuration B which used an optical fiber. Teflon provided a transmission window from 12,000 to 4000 cm^{-1} with a maximum output around 5800 cm^{-1}. The Teflon single beam intensity was decreased approximately 25% from that of without any specific absorption features. The attenuation due to scattering and refraction from Teflon itself resulted in a decrease of light throughput. The sharp intensity attenuations around 7260 and 5340 cm^{-1} were from the absorption of water vapor in air.

The intensity of the Teflon single beam collected by configuration B is significantly decreased due to the strong attenuation

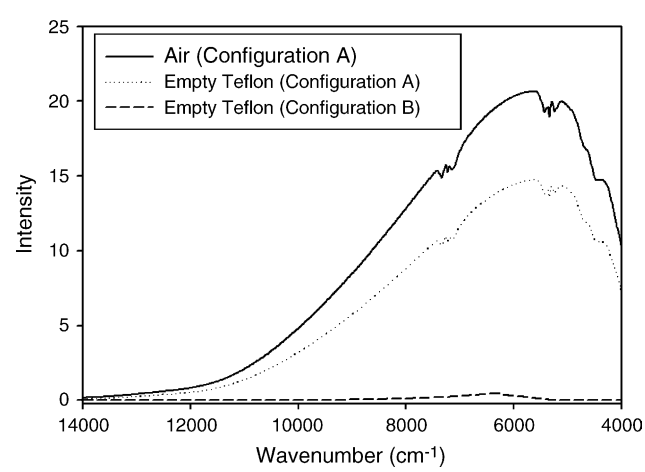

Fig. 2. The single beam spectra of air and an empty Teflon tube collected by configuration A, and the single beam spectrum of an empty Teflon tube collected by configuration B.

of the optical fiber and loss of light due to optical fiber coupling. The total extinction in the 5500–4000 cm^{-1} range originated from absorption by the optical fiber. While Teflon single beam intensities from both configuration A and B could not be compared linearly, since two different detectors were employed, the significant decrease in intensity in configuration B is obvious.

The root mean square noise was calculated to quantify the background noise levels for both configurations. Thirty zero-absorption lines (baselines) were acquired and ratioed to two consecutive Teflon single beam spectra acquired using both configuration A and B. Where applicable spectral features were present, only the 9000–7500 cm^{-1} range was used to calculate the RMS noise (reference). The baselines were straight and the least squares method successfully calculated the RMS noise over this range. The resulting averaged RMS noise levels for configuration A and B were 5.80×10^{-5} and 6.65×10^{-5} a.u., respectively. The more sensitive Ge detector in configuration B only slightly increased the RMS noise level and this allowed for reasonable quantitative calibration using this configuration. The ability to use an optical fiber for spectral collection is critical because most wet stations require remote, multi-stream analysis. In light of this, configuration B was used solely to collect spectra for the remainder of the study.

To examine the spectral difference when Teflon was used in place of quartz, NIR spectra of water were collected by using both Teflon and quartz cells. The resulting spectra are shown in Fig. 3. Baselines of both spectra were vertically adjusted for better visual comparison. The quartz cell pathlength (9.85 mm) was similar as the Teflon tube (inner diameter: 9.68 mm). While there were no significant spectral differences between the two spectra in terms of shape, the magnitude of absorption for the Teflon cell was lower. This may be explained by the circular shape of the Teflon cell as opposed to the rectangular quartz cell. The apparent beam cross-section could be slightly smaller depending on slight variations in cell positioning. This resulted in lower absorptions when Teflon was used, even though the measured pathlengths were almost similar. Consistent use of the Teflon cell did allow for the collection of repeatable NIR spectra.

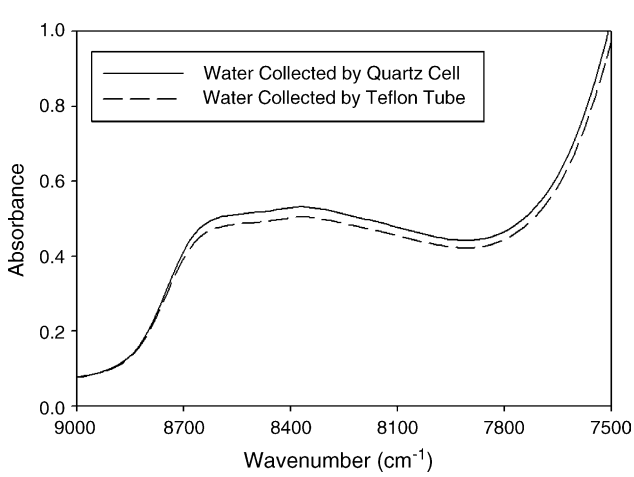

Fig. 3. NIR spectra of water by using both Teflon and quartz cells.

3.2. NIR spectral features

To investigate stepwise spectral variations, several binary mixtures of HCl and water were prepared and corresponding NIR spectra were examined. Fig. 4(a) shows the NIR spectra (9000–7500 cm^{-1} range) of pure water, 18.0, 25.0 and 32.0 wt.% HCl. Moderate and systematic spectral variations were observed with increasing HCl concentrations, even though HCl does not have NIR absorption due to its dissociation into proton and chloride ions. As previously observed, if ionic species interact with other components and change their spectral characteristics, the concentration of non-absorbing component can be indirectly measured [6–8]. In this case, water can be used as a medium for indirect measurement of HCl concentration. As shown, the water band at 8400 cm^{-1} (the combination of OH bands, e.g. $\approx$6800 cm^{-1} + 1600 cm^{-1} (bending) = 8400 cm^{-1}) is clearly decreasing with increasing HCl concentration due to water displacement in beam path by protons and chloride ions. The relative number of OH groups in the beam path is decreasing due to an increasing presence of these ions. The subsequent decrease of the water band without change of spectral features is clearly identified according to the increase of HCl concentration.

Fig. 4(b) shows NIR spectra of water and 5.0, 10.0 and 15.0 wt.% acetic acid in water. The peak variation pattern for acetic acid is obviously different from that of HCl. The overall

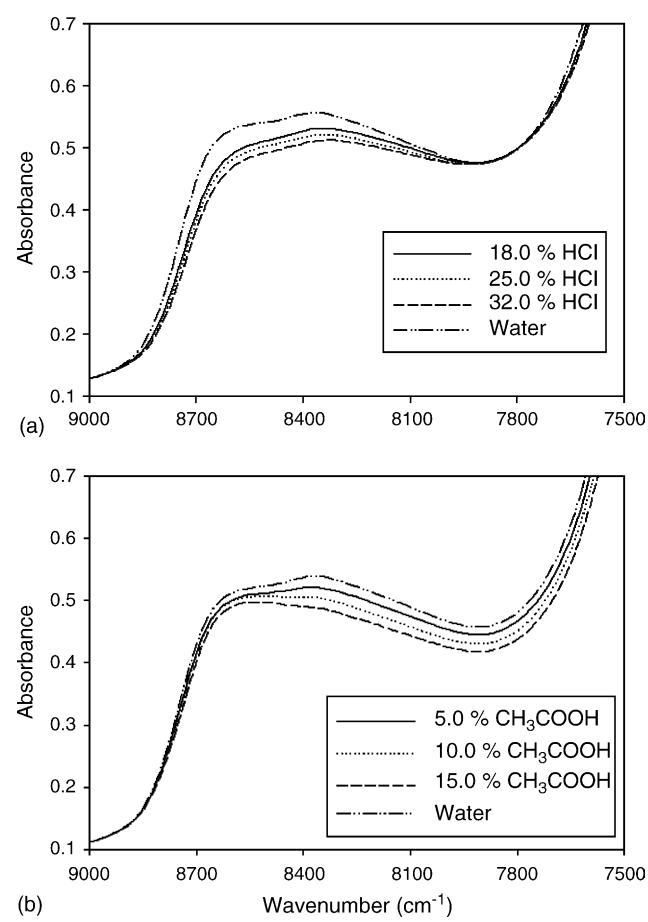

Fig. 4. NIR spectra (9000–7500 cm^{-1} range) of different concentrations of HCl (a) and acetic acid (b) in water.

absorbance decreased with increased acetic acid concentration because water has higher molar absorptivity than acetic acid. Since acetic acid has CH functional groups, a corresponding band around 8600 cm^{-1} (second overtone of the CH band, e.g. $\approx 2870\,cm^{-1} \times 3 = 8610\,cm^{-1}$) appeared and becomes more dominant with increasing acetic acid concentrations, while the OH band at 8400 cm^{-1} weakens.

Overall, the spectral variations that originated from the presence of HCl and acetic acid are clearly identified using a Teflon cell. Although spectral variations originated either from indirect water peak change or from the direct absorption of a component, unique spectral variations based on compositional change provided enough selective information to quantify each component.

3.3. PLS calibration

Because the optical quality of Teflon might not be as consistent as other materials such as quartz or CaF_2, it was necessary to address possible variations in calibration performance. To investigate the effects of optical quality on calibration performance, NIR spectra were collected using three different Teflon tubes containing one of the 28 multi-component standards. The three Teflon tubes selected were visually similar in terms of transparency, as shown in Fig. 5 (designated as Teflon 1, Teflon 2 and Teflon 3). The letters "NIR" printed on white paper were placed behind the Teflon tubes to compare transparency. Three rectangular quartz cells that had the closest pathlength of Teflon were also used to collect spectra for the same samples. For each sample, three spectra were collected consecutively without removing cells.

Partial least squares regression [9,10] was used to build calibration models for each dataset (total six sets: three sets for three Teflon tubes and three sets for three quartz cells). The 9000–7500 cm^{-1} range containing the combination OH band and the second overtone CH bands was solely selected as described. The optimum number of factors was identified as the one giving a minimum standard error of cross validation (SECV). The cross validation was used as a validation method by dividing the data set into four segments. The decreasing pattern of SECV as a function of the number of PLS factors was examined. The SECV decreased sharply for the first several factors and gradually decreased for the following factors. At a certain factor, it started to increase. The factor before the increase of SECV was chosen as the optimum number of factors [11,12].

Overall calibration results are summarized in Table 1. Numbers in parenthesis correspond to the number of factors used. The

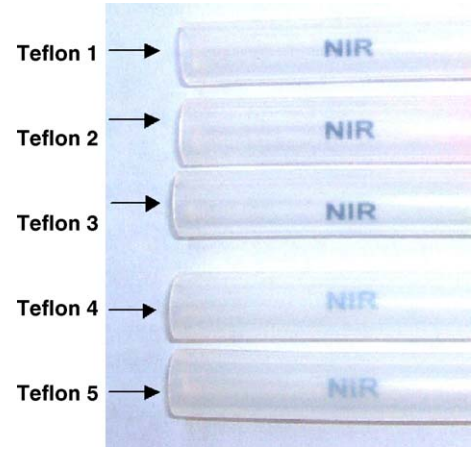

Fig. 5. Picture of five different Teflon tubes. Teflon 4 and 5 were purchased separately and are less transparent visually.

Table 1
PLS calibration results (SECV, and corresponding R^2 and slope) for three components using raw and second derivative spectra

		Teflon 1	Teflon 2	Teflon 3	Quartz 1	Quartz 2	Quartz 3
Raw spectra							
HCl	SECV (wt.%)	0.17 (6)	0.13 (6)	0.16 (6)	0.12 (6)	0.12 (6)	0.12 (6)
	R^2	0.99901	0.99944	0.99919	0.99948	0.99950	0.99951
	Slope	1.00001	1.00002	1.00000	0.99999	1.00001	1.00000
Acetic acid	SECV (wt.%)	0.25 (7)	0.20 (7)	0.20 (7)	0.05 (6)	0.06 (6)	0.05 (6)
	R^2	0.99674	0.99796	0.99779	0.99988	0.99980	0.99986
	Slope	0.99998	0.99999	0.99999	0.99999	0.99999	1.00000
Water	SECV (wt.%)	0.21 (7)	0.14 (7)	0.23 (7)	0.12 (6)	0.13 (6)	0.12 (6)
	R^2	0.99906	0.99960	0.99888	0.99966	0.99965	0.99967
	Slope	1.00002	1.00001	1.00002	1.00003	1.00000	1.00003
Second derivative spectra							
HCl	SECV (wt.%)	0.14 (6)	0.11 (6)	0.14 (6)	0.12 (6)	0.13 (6)	0.13 (6)
	R^2	0.99936	0.99957	0.99931	0.99953	0.99947	0.99947
	Slope	1.00001	1.00001	1.00002	1.00002	1.00002	1.00001
Acetic acid	SECV (wt.%)	0.14 (7)	0.11 (7)	0.10 (7)	0.04 (6)	0.04 (6)	0.05 (6)
	R^2	0.99900	0.99940	0.99951	0.99993	0.99992	0.99990
	Slope	0.99999	0.99998	0.99999	0.99998	0.99999	0.99998
Water	SECV (wt.%)	0.12 (7)	0.10 (7)	0.12 (7)	0.12 (6)	0.11 (6)	0.12 (6)
	R^2	0.99966	0.99977	0.99966	0.99969	0.99975	0.99968
	Slope	1.00001	1.00002	1.00001	1.00002	1.00001	1.00001

Numbers in parentheses correspond to the number of factors used.

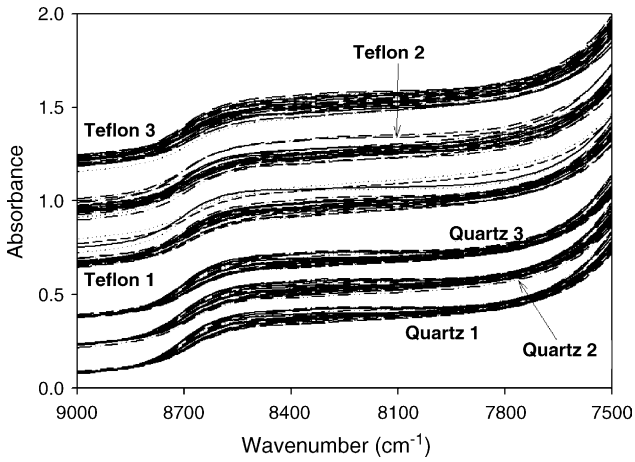

Fig. 6. Twenty-eight spectra of sample data set collected by three Teflon and three quartz cells.

SECVs from three quartz cells are clearly better than those from three Teflon cells. It was reasonable that the spectra acquired by Teflon should be less reproducible compared to the spectra acquired by quartz cell due to scattering and refraction. Fig. 6 shows 28 spectra of the sample data set collected by three Teflon and three quartz cells. The spectra of each sample set were offset for clear visual comparison. The maximum absorptions in each case were around 0.5 a.u. that were well within linear range of the detector. As noted, the baselines of spectra collected in Teflon cells varied more than those obtained in quartz cells. Additionally, the quantitative identity of a sample might not exactly be represented by the spectra collected in Teflon cells due to the deviation from Beer's law as a result of the round Teflon surface. Therefore, the SECVs from Teflon cells are higher, especially for acetic acid. Since the molar absorptivity of acetic acid is lower than that of water, the magnitude of spectral variation based on concentration change would be correspondingly smaller. Because of this, the calibration performance of acetic acid is lessened under the situation of degraded spectral reproducibility. The SECVs from the three different quartz cells were nearly identical, while those from Teflon cells varied slightly. Scatter plots showing correlation between reference and NIR analyses for both cases are shown in Fig. 7. The results using quartz cells showed better correlations, especially for acetic acid. Even though the data are more scattered when Teflon cells were used, there was still good correlation with reference data, indicat-

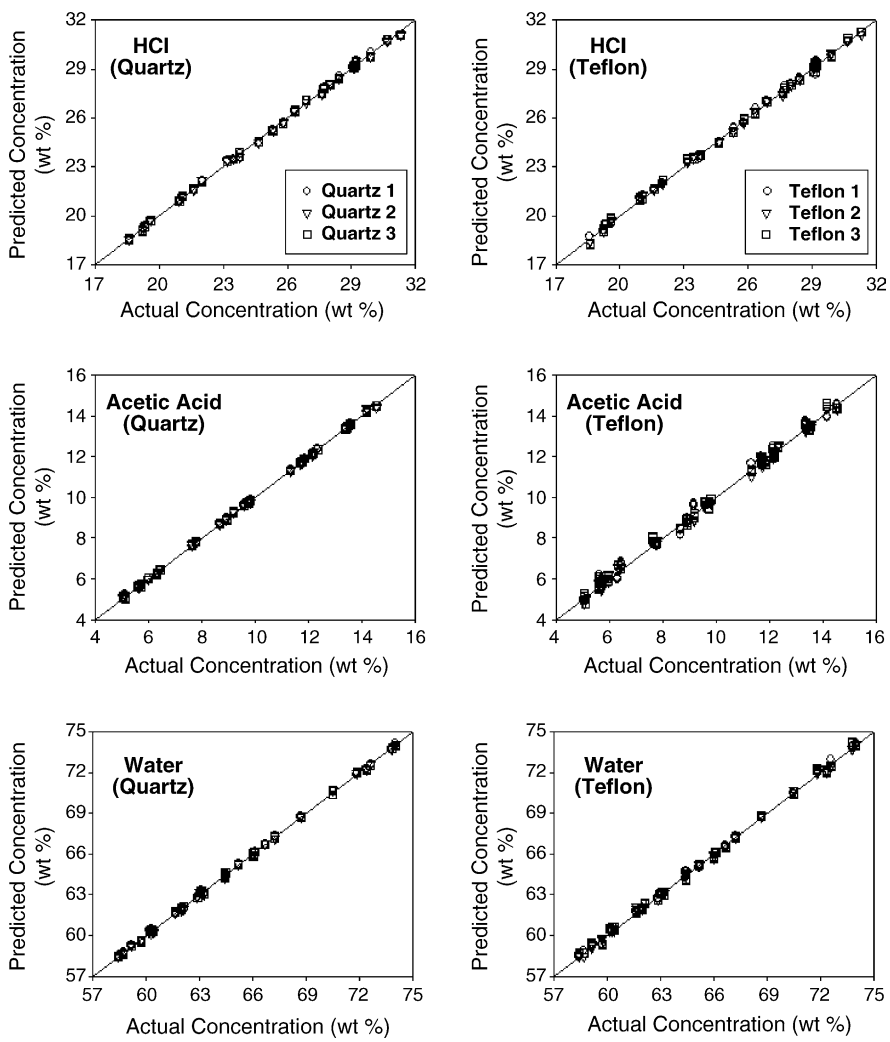

Fig. 7. Scatter plots for three components showing correlation between reference and NIR analyses for both cases (Teflon and quartz).

Table 2
The prediction results (SEP, and corresponding R^2 and slope) for three components by predicting the spectra collected with Teflon 2 and 3 using PLS model developed using the spectra collected with Teflon 1 (raw and second derivative spectra)

		Teflon 2	Teflon 3	Quartz 2	Quartz 3
Raw spectra					
HCl	SEP (wt.%)	0.16	0.18	0.13	0.14
	R^2	0.99931	0.99907	0.99946	0.99934
	Slope	0.99770	1.01113	1.00686	1.00601
Acetic acid	SEP (wt.%)	0.21	0.24	0.08	0.08
	R^2	0.99801	0.99739	0.99984	0.99975
	Slope	1.00675	0.97444	0.99722	0.99738
Water	SEP (wt.%)	0.20	0.24	0.13	0.15
	R^2	0.99912	0.99883	0.99965	0.99951
	Slope	1.00126	0.98570	1.00203	0.99958
Second derivative spectra					
HCl	SEP (wt.%)	0.24	0.25	0.13	0.15
	R^2	0.99817	0.99780	0.99945	0.99930
	Slope	0.99644	1.00075	1.00818	1.00967
Acetic acid	SEP (wt.%)	0.21	0.18	0.04	0.05
	R^2	0.99760	0.99821	0.99991	0.99987
	Slope	1.00265	1.00466	0.99963	0.99962
Water	SEP (wt.%)	0.22	0.22	0.13	0.15
	R^2	0.99897	0.99902	0.99965	0.99956
	Slope	0.98896	0.99009	1.00407	1.00644

ing that this methodology has a strong potential for applications in an industrial environment where the 0.2–0.3% level of standard error of predictions (SEPs) is required.

3.4. Prediction stability with different Teflon tubes

To have reliability of the proposed Teflon-based measurement in real industry, the prediction accuracy should be maintained even when different Teflon tubes are used. If it is not realized, calibration models must be developed for each Teflon cell used, which is practically difficult and unrealistic. To evaluate the prediction stability, the spectra collected by Teflon 2 and 3 were predicted using the PLS model developed with the spectra collected by Teflon 1. The same procedure was repeated for the quartz dataset for comparison. The prediction results are summarized in Table 2. By comparison with SECV from Teflon 1, there was no significant degradation in standard error of predictions when the spectra collected by two different Teflon tubes were predicted. This is indirect evidence that fairly reproducible spectra are acquired even though different Teflon tubes are used. At the same time, similar results were achieved for the case quartz cells.

For a more in-depth investigation, difference spectra of the samples were generated by subtracting the spectra collected with Teflon 2 and 3 from the spectrum collected with Teflon 1. This allowed for net spectroscopic variations that originate from different Teflon cells to be examined. Twenty-eight difference spectra for each set were calculated and the same procedure was repeated for the quartz data set. The resulting difference spectra were averaged and are shown in Fig. 8. As expected, the spectral differences were much smaller for quartz. For Teflon cells, the maximum magnitude of difference was approximately 0.002 a.u. Therefore, it is proposed that a calibration model can be used without any adjustment if the maximum magnitude of spectral difference is below or around 0.002 a.u. when different Teflon tubes are used.

3.5. Second derivative algorithm

As shown in Fig. 6, the baselines of spectra collected by Teflon cells vary. To reduce baseline variations present in the raw spectra, the second derivative of the spectra was used. Each of the raw spectra was converted to its second derivative and PLS regression was performed once again. Results are summarized in Tables 1 and 2, and compared with those from the use of only raw spectra. In case of Teflon, individual SECV was improved using a second derivative algorithm. No significant difference

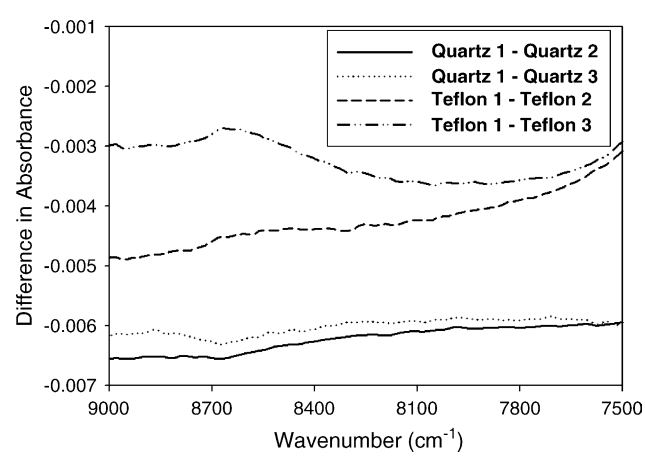

Fig. 8. The averaged difference spectra for Teflon and quartz cells. A total of 28 difference spectra for each data set were generated and then averaged.

Table 3
The SEPs (corresponding R^2 and slope) and mean percent errors (MPEs) for three components by predicting the spectra that were collected in a separate period using five different Teflon tubes

	SEP (wt.%)	R^2	Slope	MPE (%)	Repeatability (wt.%)
HCl					
Teflon 1	0.16	0.99947	0.97947	0.51	0.05
Teflon 2	0.18	0.99921	0.99407	0.62	0.07
Teflon 3	0.21	0.99903	1.03033	0.71	0.02
Teflon 4	0.21	0.99899	1.03529	0.57	0.05
Teflon 5	0.17	0.99946	0.98321	0.57	0.05
Acetic acid					
Teflon 1	0.20	0.99897	0.99110	3.00	0.09
Teflon 2	0.21	0.99838	0.97968	2.68	0.08
Teflon 3	0.19	0.99903	0.99162	2.03	0.09
Teflon 4	0.18	0.99905	1.02705	1.99	0.09
Teflon 5	0.21	0.99896	0.98224	2.36	0.10
Water					
Teflon 1	0.18	0.99960	0.99570	0.23	0.07
Teflon 2	0.18	0.99954	0.99506	0.22	0.08
Teflon 3	0.24	0.99917	1.01241	0.32	0.06
Teflon 4	0.24	0.99918	1.01222	0.28	0.08
Teflon 5	0.23	0.99950	0.98213	0.26	0.08

Repeatability corresponds to standard deviation of 20 predicted values from 20 consecutive spectra.

was observed for the case of quartz. The baseline variations from quartz cells were much smaller, so the second derivative is not necessarily required to improve calibration. When the spectra collected by Teflon 2 and 3 were predicted by the PLS model developed using the second derivative spectra of Teflon 1, the resulting SEPs were increased (Table 2) compared to corresponding SECVs (Table 1). The second derivative algorithm was successfully used to reduce baseline variations in a given Teflon set but did not effectively compensate for the variations in the different Teflon cells. The pattern of spectral differences among Teflon cells is hard to characterize and therefore, it is difficult to improve calibration performance with simple pre-processing techniques such as using the second derivative algorithm. On the contrary, the calibration accuracy was maintained for the prediction of spectra from different quartz cells. There were no significant differences between SECVs and resulting SEPs when quartz cells were used.

3.6. Prediction robustness

The prediction robustness of Teflon-based NIR measurement was evaluated by predicting the spectra of 10 independent samples that were prepared fresh specifically for these experiments. The concentrations of 10 new samples matched the concentrations of 10 samples in the original dataset. Simultaneously, 20 spectra per sample were consecutively collected without moving the Teflon cells to evaluate measurement repeatability. It took approximately 18 min to collect 20 spectra. The procedure was repeated using the same three different Teflon cells (Teflon 1, 2 and 3) as shown in Fig. 5.

To demonstrate the practical applicability of the method, variation of prediction performance using different qualities of Teflon was addressed. For this purpose, two more Teflon tubes were used that were visually different from three existing Teflon tubes. As shown in Fig. 5 (designated as Teflon 4 and 5) the two new Teflon tubes appear visually less transparent than Teflon 1, 2 and 3. The spectra of the same 10 independent samples for prediction robustness were collected using these two Teflon tubes. In total, the spectra of 10 new samples were collected using five different Teflon tubes. For each Teflon tube, 200 spectra were collected (10 samples and 20 consecutive spectra per sample). The percent variation of pathlengths of five different Teflon cells was 0.87% (standard deviation: 0.08 mm).

The overall prediction results using five Teflon tubes are summarized in Table 3. The resulting SEPs and mean percent error (MPEs) are presented. Even when two different Teflon tubes with less transparency were used, there were no significant variations in SEPs and MPEs. The resulting standard deviations of five SEP values from five different cells (Table 3) were 0.02% for HCl, 0.01% for acetic acid and 0.03% for water. The standard deviations are smaller than repeatabilities for corresponding each component; therefore, it can be concluded that approximately 1% variation of Teflon pathlength does not significantly affect prediction performance. The slight increase in MPE variability for acetic acid is due to its relatively low concentration range. Difference spectra were generated as described previously and results are shown in Fig. 9. The spectra from Teflon 1 to 5 were subtracted from the original spectra used for PLS model development using Teflon 1. The maximum magnitudes of difference were increased up to 0.003–0.004 a.u.; however, these differences were still small and resulting SEPs were not significantly influenced.

The repeatability of the Teflon-based NIR measurement was evaluated by calculating standard deviations of 20 predicted values from 20 consecutive spectra of each sample collected over 18 min. The averaged standard deviations for each Teflon tube are also shown in Table 3. Highly repeatable prediction results for the three components were achieved.

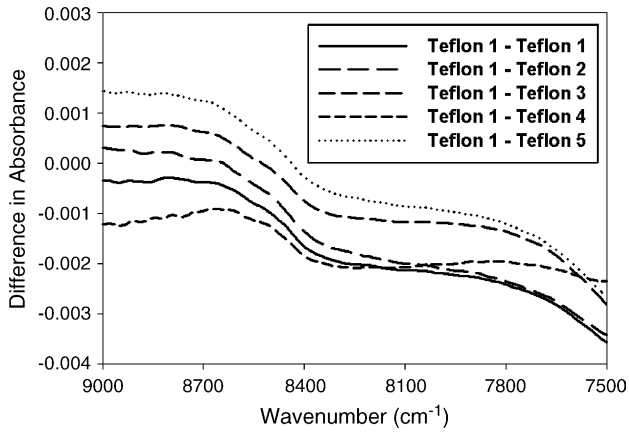

Fig. 9. The averaged difference spectra of five different Teflon tubes.

4. Conclusions

The overall results present the utility of Teflon-based NIR for the monitoring of etching solutions in electronic industries. It can also be successfully integrated into processes for use as an on-line and remote monitoring analytical tool. The prediction results were reproducible even when Teflon tubes were changed and the repeatability was acceptable. Before this methodology can be confidently adopted in the field, several analytical issues should be addressed. First, spectral differences amongst a larger number of Teflon tubes should be characterized. Second, the relationship between calibration performance and temperature variation should be investigated. Variations in temperature could affect calibration performance, especially for solutions containing a temperature sensitive sample [13,14]. In addition, the effect of the presence of numerous bubbles in the sample on NIR measurements should be quantified. Currently, a pilot-scale circulation system that closely simulates an actual process with temperature control capability is being built. Future research will use this system to address the proposed scientific hurdles and also develop spectral normalization method to correct possible pathlength variation of Teflon tube.

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